Supply Chain Analytics — SQL Queries & Business Insights

DataCo Global Supply Chain Dataset | Python + SQL project

Project context

This document contains the SQL business questions, queries, and concise interpretations extracted and 整理 ed from the project report. The dataset contains 180,519 records, 65,752 unique orders, and 20,652 unique customers, covering five markets. The SQL examples use the table name supply_chain_orders.

Before running the queries, load the dataset into a table named supply_chain_orders and check that the column names match. Most queries use MySQL 8.0 syntax. Query outputs depend on the loaded data and database schema.

Main business problems covered

- Late deliveries and differences between scheduled and actual shipping times.
- Negative-profit orders and possible margin erosion from discounting.
- Suspected-fraud records and unusual profit patterns that need investigation.
- Differences in revenue and profitability across markets, regions, products, departments, and customer segments.
- Order cancellations, on-hold orders, and payment/order-processing bottlenecks.
- Seasonality in sales and the need to plan inventory and logistics capacity.
- Product revenue concentration and customer lifetime value/retention.

Level 1 — Basic SQL

1. What is the total revenue across all order items?

Business question: What is the total revenue across all order items?

SQL

```
SELECT SUM(Sales) AS total_revenue
FROM supply_chain_orders;
```

Answer / business interpretation: The report lists total revenue of approximately $36.78 million.

2. How many unique orders are in the dataset?

Business question: How many unique orders are in the dataset?

SQL

```
SELECT COUNT(DISTINCT Order_Id) AS unique_orders
FROM supply_chain_orders;
```

Answer / business interpretation: The dataset contains 65,752 unique orders.

3. What delivery statuses appear in the data?

Business question: What delivery statuses appear in the data?

SQL

```
SELECT DISTINCT Delivery_Status
FROM supply_chain_orders
ORDER BY Delivery_Status;
```

Answer / business interpretation: The report identifies four main delivery outcomes: late delivery, on-time shipping, advance shipping, and cancellation.

4. Which 10 order records have the highest profit?

Business question: Which 10 order records have the highest profit?

SQL

```
SELECT Order_Id, Sales, Order_Profit_Per_Order
FROM supply_chain_orders
ORDER BY Order_Profit_Per_Order DESC
LIMIT 10;
```

Answer / business interpretation: This query identifies the most profitable order records for further review.

5. How many unique orders are associated with each market?

Business question: How many unique orders are associated with each market?

SQL

```
SELECT Market, COUNT(DISTINCT Order_Id) AS order_count
FROM supply_chain_orders
GROUP BY Market
ORDER BY order_count DESC;
```

Answer / business interpretation: LATAM is described as the largest market by order/revenue volume; Africa is the smallest market.

6. How frequently is each shipping mode used?

Business question: How frequently is each shipping mode used?

SQL

```
SELECT Shipping_Mode, COUNT(*) AS record_count
FROM supply_chain_orders
GROUP BY Shipping_Mode
ORDER BY record_count DESC;
```

Answer / business interpretation: Standard Class accounts for roughly 60% of order records, making its performance operationally important.

7. How many unique customers belong to each segment?

Business question: How many unique customers belong to each segment?

SQL

```
SELECT Customer_Segment,
       COUNT(DISTINCT Customer_Id) AS customer_count
FROM supply_chain_orders
GROUP BY Customer_Segment
ORDER BY customer_count DESC;
```

Answer / business interpretation: The dataset has Consumer, Corporate, and Home Office customer segments.

8. What is the average sales value by payment type?

Business question: What is the average sales value by payment type?

SQL

```
SELECT Type AS payment_type,
       ROUND(AVG(Sales), 2) AS average_sales
FROM supply_chain_orders
GROUP BY Type
ORDER BY average_sales DESC;
```

Answer / business interpretation: Average sales values are similar across payment types, around $200–$210 according to the report.

9. What share of records uses each payment type?

Business question: What share of records uses each payment type?

SQL

```
SELECT Type AS payment_type,
       COUNT(*) AS record_count,
       ROUND(100.0 * COUNT(*) / (SELECT COUNT(*) FROM supply_chain_orders), 1) AS pct
FROM supply_chain_orders
GROUP BY Type
ORDER BY record_count DESC;
```

Answer / business interpretation: Debit is reported as the most common payment type (about 38%); cash is the least common (about 11%).

Level 2 — Intermediate SQL

10. What is the late-delivery rate by shipping mode?

Business question: What is the late-delivery rate by shipping mode?

SQL

```
SELECT Shipping_Mode,
       COUNT(*) AS total_records,
       SUM(CASE WHEN Delivery_Status = 'Late delivery' THEN 1 ELSE 0 END) AS late_records,
       ROUND(100.0 * SUM(CASE WHEN Delivery_Status = 'Late delivery' THEN 1 ELSE 0 END)
             / COUNT(*), 1) AS late_pct
FROM supply_chain_orders
GROUP BY Shipping_Mode
ORDER BY late_pct DESC;
```

Answer / business interpretation: The report flags poor on-time performance in premium shipping modes; audit the service levels and compare the rate with customer promises.

11. How does revenue change month by month?

Business question: How does revenue change month by month?

SQL

```
SELECT DATE_FORMAT(order_date, '%Y-%m') AS month,
       ROUND(SUM(Sales) / 1000000, 2) AS revenue_millions
FROM supply_chain_orders
GROUP BY DATE_FORMAT(order_date, '%Y-%m')
ORDER BY month;
```

Answer / business interpretation: The report identifies recurring fourth-quarter sales peaks, useful for inventory and carrier-capacity planning.

12. Which customer segment contributes the most profit?

Business question: Which customer segment contributes the most profit?

SQL

```
SELECT Customer_Segment,
       ROUND(SUM(Order_Profit_Per_Order), 0) AS total_profit,
       ROUND(AVG(Order_Profit_Per_Order), 2) AS average_profit
FROM supply_chain_orders
GROUP BY Customer_Segment
ORDER BY total_profit DESC;
```

Answer / business interpretation: The Consumer segment contributes the highest total profit in the report, while segment-level averages support differentiated strategies.

13. Which suspected-fraud orders also have negative profit?

Business question: Which suspected-fraud orders also have negative profit?

SQL

```
SELECT Order_Id, Sales, Order_Profit_Per_Order, Order_Status
FROM supply_chain_orders
WHERE Order_Profit_Per_Order < 0
  AND UPPER(Order_Status) = 'SUSPECTED_FRAUD'
ORDER BY Order_Profit_Per_Order ASC
LIMIT 20;
```

Answer / business interpretation: This returns records requiring investigation; negative profit alone is not proof of fraud.

14. Which 10 products generate the most revenue?

Business question: Which 10 products generate the most revenue?

SQL

```
SELECT Product_Name,
       ROUND(SUM(Sales) / 1000, 1) AS revenue_thousands,
       COUNT(DISTINCT Order_Id) AS order_count
FROM supply_chain_orders
GROUP BY Product_Name
ORDER BY SUM(Sales) DESC
LIMIT 10;
```

Answer / business interpretation: The report finds revenue concentrated in a small group of products; disruptions to top products could have an outsized effect.

15. How does average profit vary across discount bands?

Business question: How does average profit vary across discount bands?

SQL

```
SELECT CASE
         WHEN Order_Item_Discount_Rate = 0 THEN 'No Discount'
         WHEN Order_Item_Discount_Rate <= 0.10 THEN 'Low'
         WHEN Order_Item_Discount_Rate <= 0.20 THEN 'Medium'
         ELSE 'High'
       END AS discount_band,
       ROUND(AVG(Order_Profit_Per_Order), 2) AS average_profit
FROM supply_chain_orders
GROUP BY discount_band
ORDER BY average_profit DESC;
```

Answer / business interpretation: The report describes a negative relationship between discount depth and profit, with high discounts associated with weaker margins.

16. How do sales and profit vary by year and quarter?

Business question: How do sales and profit vary by year and quarter?

SQL

```
SELECT YEAR(order_date) AS order_year,
       QUARTER(order_date) AS order_quarter,
       ROUND(SUM(Sales) / 1000000, 2) AS sales_millions,
       ROUND(SUM(Order_Profit_Per_Order) / 1000, 1) AS profit_thousands
FROM supply_chain_orders
GROUP BY YEAR(order_date), QUARTER(order_date)
ORDER BY order_year, order_quarter;
```

Answer / business interpretation: The report identifies Q4 as a recurring strong quarter for sales and profit.

17. Which regions have the highest average shipping delay?

Business question: Which regions have the highest average shipping delay?

SQL

```
SELECT Order_Region,
       ROUND(AVG(Days_for_shipping_real - Days_for_shipment_scheduled), 2) AS avg_delay_days
FROM supply_chain_orders
GROUP BY Order_Region
ORDER BY avg_delay_days DESC
LIMIT 10;
```

Answer / business interpretation: Use this ranking to prioritise regions for logistics review; positive values indicate actual shipping took longer than scheduled.

18. Which product categories generate the most revenue?

Business question: Which product categories generate the most revenue?

SQL

```
SELECT Category_Name,
       COUNT(DISTINCT Order_Id) AS order_count,
       ROUND(SUM(Sales) / 1000, 1) AS revenue_thousands
FROM supply_chain_orders
GROUP BY Category_Name
ORDER BY SUM(Sales) DESC
LIMIT 15;
```

Answer / business interpretation: Category-level revenue helps identify the product groups that drive sales and deserve closer inventory and margin monitoring.

19. Which markets have the highest share of cancelled or on-hold records?

Business question: Which markets have the highest share of cancelled or on-hold records?

SQL

```
SELECT Market,
       COUNT(*) AS total_records,
       SUM(CASE WHEN UPPER(Order_Status) IN ('CANCELED', 'ON_HOLD') THEN 1 ELSE 0 END) AS
problem_records,
       ROUND(100.0 * SUM(CASE WHEN UPPER(Order_Status) IN ('CANCELED', 'ON_HOLD') THEN 1 ELSE 0 END)
             / COUNT(*), 1) AS problem_pct
FROM supply_chain_orders
GROUP BY Market
ORDER BY problem_pct DESC;
```

Answer / business interpretation: The result highlights where order-processing or payment bottlenecks may need further investigation.

20. How do departments compare on revenue, profit, and margin?

Business question: How do departments compare on revenue, profit, and margin?

SQL

```
SELECT Department_Name,
       COUNT(DISTINCT Order_Id) AS order_count,
       ROUND(SUM(Sales) / 1000000, 2) AS revenue_millions,
       ROUND(SUM(Order_Profit_Per_Order) / 1000, 1) AS profit_thousands,
       ROUND(100.0 * SUM(Order_Profit_Per_Order) / NULLIF(SUM(Sales), 0), 1) AS margin_pct
FROM supply_chain_orders
GROUP BY Department_Name
ORDER BY SUM(Sales) DESC;
```

Answer / business interpretation: Comparing revenue with margin helps distinguish high-volume departments from those that are more profitable.

21. What are the average item quantity and item value by customer segment?

Business question: What are the average item quantity and item value by customer segment?

SQL

```
SELECT Customer_Segment,
       ROUND(AVG(Order_Item_Quantity), 2) AS avg_quantity,
       ROUND(AVG(Order_Item_Total), 2) AS avg_item_value
FROM supply_chain_orders
GROUP BY Customer_Segment
ORDER BY avg_item_value DESC;
```

Answer / business interpretation: The report describes Home Office as a smaller segment with relatively high average order-item value.

22. How do actual shipping days compare with scheduled days by shipping mode?

Business question: How do actual shipping days compare with scheduled days by shipping mode?

SQL

```
SELECT Shipping_Mode,
       ROUND(AVG(Days_for_shipment_scheduled), 1) AS scheduled_days,
       ROUND(AVG(Days_for_shipping_real), 1) AS actual_days,
       ROUND(AVG(Days_for_shipping_real - Days_for_shipment_scheduled), 2) AS avg_delay_days
FROM supply_chain_orders
GROUP BY Shipping_Mode
ORDER BY avg_delay_days DESC;
```

Answer / business interpretation: Positive average delay indicates that the mode is taking longer than scheduled; use this to review service-level commitments.

23. Which order statuses have the highest average sales value?

Business question: Which order statuses have the highest average sales value?

SQL

```
SELECT Order_Status,
       COUNT(DISTINCT Order_Id) AS order_count,
       ROUND(AVG(Sales), 2) AS avg_sales
FROM supply_chain_orders
GROUP BY Order_Status
ORDER BY avg_sales DESC;
```

Answer / business interpretation: This compares the value of orders at different processing stages and can help prioritise unresolved high-value orders.

Level 3 — Advanced SQL (CTEs and Window Functions)

24. What is the running total of monthly revenue?

Business question: What is the running total of monthly revenue?

SQL

```
WITH monthly AS (
    SELECT DATE_FORMAT(order_date, '%Y-%m') AS month,
           SUM(Sales) AS monthly_revenue
    FROM supply_chain_orders
    GROUP BY DATE_FORMAT(order_date, '%Y-%m')
)
SELECT month, monthly_revenue,
       SUM(monthly_revenue) OVER (ORDER BY month) AS running_revenue
FROM monthly
```

```
ORDER BY month;
```

Answer / business interpretation: A running total shows cumulative revenue over time and supports performance tracking.

25. What are the top three products by revenue within each category?

Business question: What are the top three products by revenue within each category?

SQL

```
WITH product_sales AS (
    SELECT Category_Name, Product_Name, SUM(Sales) AS total_sales
    FROM supply_chain_orders
    GROUP BY Category_Name, Product_Name
),
ranked AS (
    SELECT Category_Name, Product_Name, total_sales,
           RANK() OVER (PARTITION BY Category_Name ORDER BY total_sales DESC) AS category_rank
    FROM product_sales
)
SELECT Category_Name, Product_Name, total_sales, category_rank
FROM ranked
WHERE category_rank <= 3
ORDER BY Category_Name, category_rank;
```

Answer / business interpretation: This highlights leading products within each category rather than only across the entire business.

26. What is month-over-month revenue growth?

Business question: What is month-over-month revenue growth?

SQL

```
WITH monthly AS (
    SELECT DATE_FORMAT(order_date, '%Y-%m') AS month, SUM(Sales) AS revenue
    FROM supply_chain_orders
    GROUP BY DATE_FORMAT(order_date, '%Y-%m')
),
with_previous AS (
    SELECT month, revenue, LAG(revenue) OVER (ORDER BY month) AS previous_revenue
    FROM monthly
)
SELECT month, revenue, previous_revenue,
       ROUND(100.0 * (revenue - previous_revenue) / NULLIF(previous_revenue, 0), 1) AS mom_growth_pct
FROM with_previous
ORDER BY month;
```

Answer / business interpretation: The growth rate distinguishes genuine month-to-month changes from changes in absolute revenue.

27. How can customers be grouped by lifetime spend?

Business question: How can customers be grouped by lifetime spend?

SQL

```
WITH clv AS (
    SELECT Customer_Id,
           SUM(Sales) AS total_spent,
           COUNT(DISTINCT Order_Id) AS order_count
    FROM supply_chain_orders
    GROUP BY Customer_Id
)
SELECT Customer_Id, total_spent, order_count,
       NTILE(4) OVER (ORDER BY total_spent DESC) AS spend_quartile
FROM clv
```

```
ORDER BY total_spent DESC;
```

Answer / business interpretation: Quartiles help separate high-spending customers from lower-spending groups for retention and service planning.

28. What share of revenue comes from late versus non-late deliveries?

Business question: What share of revenue comes from late versus non-late deliveries?

SQL

```
WITH delivery_perf AS (
    SELECT CASE
             WHEN Days_for_shipping_real > Days_for_shipment_scheduled THEN 'Late'
             ELSE 'On Time / Early'
           END AS delivery_flag,
           SUM(Sales) AS revenue,
           AVG(Order_Profit_Per_Order) AS avg_profit
    FROM supply_chain_orders
    GROUP BY CASE
               WHEN Days_for_shipping_real > Days_for_shipment_scheduled THEN 'Late'
               ELSE 'On Time / Early'
             END
)
SELECT delivery_flag, revenue, avg_profit,
       ROUND(100.0 * revenue / SUM(revenue) OVER (), 1) AS revenue_share_pct
FROM delivery_perf;
```

Answer / business interpretation: This estimates the revenue associated with late deliveries and supports prioritisation of delivery improvements.

29. Which five regions have the highest profit margin?

Business question: Which five regions have the highest profit margin?

SQL

```
WITH region_kpi AS (
    SELECT Order_Region,
           SUM(Sales) AS total_sales,
           SUM(Order_Profit_Per_Order) AS total_profit,
           100.0 * SUM(Order_Profit_Per_Order) / NULLIF(SUM(Sales), 0) AS profit_margin_pct
    FROM supply_chain_orders
    GROUP BY Order_Region
)
SELECT Order_Region, total_sales, total_profit,
       ROUND(profit_margin_pct, 2) AS profit_margin_pct,
       RANK() OVER (ORDER BY profit_margin_pct DESC) AS margin_rank
FROM region_kpi
ORDER BY margin_rank
LIMIT 5;
```

Answer / business interpretation: The report identifies Europe as having the highest overall market profit; this query ranks individual order regions by margin.

30. Which orders have unusually low profit relative to their market?

Business question: Which orders have unusually low profit relative to their market?

SQL

```
WITH market_stats AS (
    SELECT Market,
           AVG(Order_Profit_Per_Order) AS avg_profit,
           STDDEV_POP(Order_Profit_Per_Order) AS sd_profit
    FROM supply_chain_orders
    GROUP BY Market
```

```
)
SELECT o.Order_Id, o.Customer_Id, o.Market, o.Sales,
       o.Order_Profit_Per_Order, o.Order_Status,
       ROUND(100.0 * o.Order_Profit_Per_Order / NULLIF(o.Sales, 0), 1) AS margin_pct
FROM supply_chain_orders o
JOIN market_stats m ON o.Market = m.Market
WHERE o.Order_Profit_Per_Order < m.avg_profit - 2 * m.sd_profit
ORDER BY o.Order_Profit_Per_Order
LIMIT 20;
```

Answer / business interpretation: This flags statistical outliers for review; unusual profit is a screening signal, not proof of fraud.

31. How many customers remain active in later years after their first order year?

Business question: How many customers remain active in later years after their first order year?

SQL

```
WITH first_order AS (
    SELECT Customer_Id, MIN(YEAR(order_date)) AS first_year
    FROM supply_chain_orders
    GROUP BY Customer_Id
),
cohort AS (
    SELECT f.first_year, YEAR(o.order_date) AS order_year,
           COUNT(DISTINCT o.Customer_Id) AS active_customers
    FROM supply_chain_orders o
    JOIN first_order f ON o.Customer_Id = f.Customer_Id
    GROUP BY f.first_year, YEAR(o.order_date)
)
SELECT first_year, order_year,
       order_year - first_year AS years_since_first_order,
       active_customers
FROM cohort
ORDER BY first_year, order_year;
```

Answer / business interpretation: This cohort-style view compares customer activity over time; it is a basic retention view rather than a complete retention percentage calculation.

32. How can products be classified using cumulative revenue (ABC analysis)?

Business question: How can products be classified using cumulative revenue (ABC analysis)?

SQL

```
WITH product_rev AS (
    SELECT Product_Name, SUM(Sales) AS revenue
    FROM supply_chain_orders
    GROUP BY Product_Name
),
cumulative AS (
    SELECT Product_Name, revenue,
           SUM(revenue) OVER (ORDER BY revenue DESC) AS cumulative_revenue,
           SUM(revenue) OVER () AS total_revenue
    FROM product_rev
)
SELECT Product_Name, revenue,
       ROUND(100.0 * cumulative_revenue / NULLIF(total_revenue, 0), 1) AS cumulative_pct,
       CASE
         WHEN cumulative_revenue / NULLIF(total_revenue, 0) <= 0.80 THEN 'A - High Value'
         WHEN cumulative_revenue / NULLIF(total_revenue, 0) <= 0.95 THEN 'B - Mid Value'
         ELSE 'C - Low Value'
       END AS abc_class
FROM cumulative
```

```
ORDER BY revenue DESC;
```

Answer / business interpretation: ABC analysis helps prioritise high-revenue products for inventory availability and supplier planning.

33. What is year-over-year sales growth by market?

Business question: What is year-over-year sales growth by market?

SQL

```
WITH yearly AS (
    SELECT Market, YEAR(order_date) AS yr, SUM(Sales) AS revenue
    FROM supply_chain_orders
    GROUP BY Market, YEAR(order_date)
),
with_previous AS (
    SELECT Market, yr, revenue,
           LAG(revenue) OVER (PARTITION BY Market ORDER BY yr) AS previous_year_revenue
    FROM yearly
)
SELECT Market, yr, revenue, previous_year_revenue,
       ROUND(100.0 * (revenue - previous_year_revenue) / NULLIF(previous_year_revenue, 0), 1) AS
yoy_growth_pct
FROM with_previous
ORDER BY Market, yr;
```

Answer / business interpretation: This compares market growth across years and can help guide regional investment.

34. Who are the top customers by lifetime spend?

Business question: Who are the top customers by lifetime spend?

SQL

```
WITH customer_spend AS (
    SELECT Customer_Id,
           CONCAT(MAX(Customer_Fname), ' ', MAX(Customer_Lname)) AS customer_name,
           MAX(Customer_Segment) AS customer_segment,
           MAX(Market) AS market,
           SUM(Sales) AS lifetime_value
    FROM supply_chain_orders
    GROUP BY Customer_Id
)
SELECT Customer_Id, customer_name, customer_segment, market,
       ROUND(lifetime_value, 2) AS lifetime_value,
       DENSE_RANK() OVER (ORDER BY lifetime_value DESC) AS value_rank
FROM customer_spend
ORDER BY value_rank
LIMIT 20;
```

Answer / business interpretation: This identifies high-value customers who may merit targeted retention or account-management activity.

35. What is the profit distribution by customer segment?

Business question: What is the profit distribution by customer segment?

SQL

```
SELECT Customer_Segment,
       MIN(Order_Profit_Per_Order) AS min_profit,
       AVG(Order_Profit_Per_Order) AS avg_profit,
       MAX(Order_Profit_Per_Order) AS max_profit
FROM supply_chain_orders
GROUP BY Customer_Segment;
```

Answer / business interpretation: The query compares the range and average of order profit by segment. The original report also proposed percentile values; percentile syntax varies by database.

36. What is the rolling three-month average revenue?

Business question: What is the rolling three-month average revenue?

SQL

```
WITH monthly AS (
    SELECT DATE_FORMAT(order_date, '%Y-%m') AS month, SUM(Sales) AS revenue
    FROM supply_chain_orders
    GROUP BY DATE_FORMAT(order_date, '%Y-%m')
)
SELECT month, revenue,
       ROUND(AVG(revenue) OVER (
           ORDER BY month ROWS BETWEEN 2 PRECEDING AND CURRENT ROW
       ), 0) AS rolling_3_month_avg
FROM monthly
ORDER BY month;
```

Answer / business interpretation: The rolling average smooths monthly fluctuations and makes underlying revenue trends easier to assess.

37. What are each customer's first and last order dates?

Business question: What are each customer's first and last order dates?

SQL

```
SELECT Customer_Id,
       MIN(order_date) AS first_order_date,
       MAX(order_date) AS last_order_date,
       DATEDIFF(MAX(order_date), MIN(order_date)) AS active_days,
       COUNT(DISTINCT Order_Id) AS total_orders
FROM supply_chain_orders
GROUP BY Customer_Id
HAVING COUNT(DISTINCT Order_Id) > 1
ORDER BY active_days DESC
LIMIT 20;
```

Answer / business interpretation: This identifies repeat customers and the span of their observed activity.

38. What share of total profit comes from each market?

Business question: What share of total profit comes from each market?

SQL

```
WITH market_profit AS (
    SELECT Market, SUM(Order_Profit_Per_Order) AS total_profit
    FROM supply_chain_orders
    GROUP BY Market
)
SELECT Market, total_profit,
       ROUND(100.0 * total_profit / NULLIF(SUM(total_profit) OVER (), 0), 1) AS profit_share_pct,
       RANK() OVER (ORDER BY total_profit DESC) AS profit_rank
FROM market_profit
ORDER BY profit_rank;
```

Answer / business interpretation: The report states that Europe generates the highest total profit despite being second in sales, while LATAM's profit is lower relative to its sales.

Key findings reported in the source document

- Total revenue: approximately $36.78 million.
- Total profit: approximately $3.97 million; reported overall margin: about 10.8%.
- Late-delivery rate: about 57.3% (the report also shows a 54.8% delivery-status distribution figure in one chart; metric definitions differ).
- Negative-profit records: about 18.7%.
- Suspected-fraud rate: about 2.25%; the report estimates potential exposure above $830,000. This is an estimate, not verified recovered revenue or confirmed loss.
- The report identifies recurring Q4 sales peaks and revenue concentration among a relatively small group of products.
- Europe is described as the highest-profit market, while LATAM leads sales/revenue in the report.

Notes

The SQL questions and query logic are based on the supplied report. A few queries were adjusted for clarity or MySQL 8.0 compatibility (for example, ranking results are filtered in an outer query, and the unusual-profit query calculates market statistics in a CTE). Verify all results against the actual dataset before presenting them as final business metrics.

Total included queries: 38